

The inverse of a matrix. $A \in M_m(\mathbb{R})$.

$$[A : I_m] \xrightarrow[\text{operations}]{\text{elementary}} [I_m : A^{-1}] \quad (! \text{ no column operations})$$

If $X = (v_1, \dots, v_m)$ we say that the dimension of $\langle X \rangle$ is given by the rank of the matrix

the basis is given by the

non-zero rows of the matrix.

④ $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$, $A^{-1} = ?$

! Check that $\exists A^{-1}$.

$$\begin{pmatrix} 1 & -4 \\ -2 & -4 \\ -2 & -4 \end{pmatrix}$$

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & -2 & 0 & 1 & 0 \\ 2 & -2 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & -3 & -6 & -2 & 1 & 0 \\ 0 & -6 & -3 & -2 & 0 & 1 \end{array} \right) \\ & \xrightarrow[R_3 = -\frac{1}{3}R_3]{R_2 = -\frac{1}{3}R_2} \left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 2 & 1 & \frac{2}{3} & 0 & -\frac{1}{3} \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - 2R_2]{R_1 \leftarrow R_1 - 2R_2} \\ & = \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 - \frac{4}{3} & -\frac{2}{3} & 0 \\ 0 & 1 & 2 & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & -3 & -\frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \end{array} \right) \xrightarrow{R_3 = -\frac{1}{3}R_3} \\ & = \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & -\frac{1}{3} & -\frac{2}{3} & 0 \\ 0 & 1 & 2 & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & \frac{2}{9} & -\frac{2}{9} & \frac{1}{9} \end{array} \right) \xrightarrow[R_2 \leftarrow R_2 - 2R_3]{R_1 \leftarrow R_1 + 2R_3} \\ & = \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{9} & -\frac{2}{9} & \frac{2}{9} \\ 0 & 1 & 0 & \frac{2}{9} & -\frac{1}{9} & \frac{2}{9} \\ 0 & 0 & 1 & \frac{2}{9} & -\frac{2}{9} & \frac{1}{9} \end{array} \right) \\ & \quad \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{I_3} \quad \underbrace{\begin{pmatrix} \frac{1}{9} & -\frac{2}{9} & \frac{2}{9} \\ \frac{2}{9} & -\frac{1}{9} & \frac{2}{9} \\ \frac{2}{9} & -\frac{2}{9} & \frac{1}{9} \end{pmatrix}}_{A^{-1}} \end{aligned}$$

! Certain method for the next exam

⑥

$$B = (e_1, e_2, e_3, e_4)$$

$$X = (v_1, v_2, v_3)$$

$$v_1 = 3e_1 + 2e_2 - 5e_3 + 4e_4$$

$$v_2 = 3e_1 - e_2 + 3e_3 - 3e_4$$

$$v_3 = 3e_1 + 5e_2 - 13e_3 + 11e_4$$

$$[X]B = ?$$

Det. its echelon form.

X is lin. indep.?

$$\begin{aligned} l_2 - l_1 \\ l_3 - l_1 \end{aligned}$$

$$X = \begin{pmatrix} 3 & 2 & -5 & 4 \\ 3 & -1 & 3 & -3 \\ 3 & 5 & -13 & 11 \end{pmatrix} \xrightarrow{l_2 \leftarrow l_2 - l_1} \begin{pmatrix} 3 & 2 & -5 & 4 \\ 0 & -3 & -8 & -7 \\ 3 & 5 & -13 & 11 \end{pmatrix} =$$

$$\xrightarrow{l_3 \leftarrow l_3 - l_1} \begin{pmatrix} 3 & 2 & -5 & 4 \\ 0 & -3 & -8 & -7 \\ 0 & 3 & -8 & 7 \end{pmatrix} \xrightarrow{l_3 \leftarrow l_3 + l_2} \begin{pmatrix} 3 & 2 & -5 & 4 \\ 0 & -3 & -8 & -7 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow v_1, v_2, v_3$ lin. dep.
 v_1, v_2 lin. indep.

$$\Rightarrow \dim X = 2.$$

$$\Rightarrow \langle X \rangle = \langle v_1, v_2 \rangle. \text{ or }$$

⑦

$$X = (v_1, v_2, v_3, v_4)$$

$$v_1 = (1, 0, 4)$$

$$v_2 = (2, 1, 0)$$

$$v_3 = (1, 5, -36)$$

$$v_4 = (2, 10, -72)$$

$$\Rightarrow X = \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 5 & -36 \\ 2 & 10 & -72 \end{pmatrix} \xrightarrow{l_4 \leftarrow l_4 - 2l_3}$$

$$= \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 5 & -36 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{l_3 \leftarrow l_3 - l_1}$$

Det. a basis and dim for $\langle X \rangle$?

$\Rightarrow v_1, v_2, v_3$ lin. indep.

$$\Rightarrow X = \langle (1, 0, 4), (2, 1, 0), (1, 5, -36) \rangle$$

$$\dim \langle X \rangle = 3.$$

$$\begin{cases} l_3 = l_3 - 9l_1 + 5l_2 \\ l_4 = l_4 - 18l_1 + 10l_2 \\ l_2 = l_2 - 2l_1 \end{cases}$$

$$\Rightarrow \langle (1, 0, 4), (0, 1, -8) \rangle.$$

$$\begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 0 & -5 & -40 \\ 0 & 0 & 0 \end{pmatrix} \dots$$

③

⑧ $X = \langle v_1, v_2, v_3 \rangle$
 $v_1 = (1, 0, 4, 3)$
 $v_2 = (0, 2, 3, 1)$
 $v_3 = (0, 4, 6, 2)$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 4 & 3 \\ 0 & 2 & 3 & 1 \\ 0 & 4 & 6 & 2 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 0 & 4 & 3 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \langle \cancel{v_3}, (1, 0, 4, 3), (0, 2, 3, 1) \rangle$$

$$\Rightarrow \dim \langle X \rangle = 2.$$

basis?

$$\dim \langle X \rangle = ?$$

⑨ $S = \langle (1, 0, 4), (2, 1, 0), (1, 1, -4) \rangle.$

$$T = \langle (-3, -2, 4), (5, 2, 4), (-2, 0, -8) \rangle$$

Det. a basis for $S, T, S+T$.

$$\underline{\dim S, T, S+T, S \cap T}$$

$$S: \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 1 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 1 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 1 & 1 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 0 & -8 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 0 & 1 & -8 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 0 & 0 & -8 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 0 & -8 \end{pmatrix}$$

$$\Rightarrow S = \langle (1, 0, 4), (0, 1, -8) \rangle. \quad \Rightarrow \dim \langle S \rangle = 2.$$

$$T: \begin{pmatrix} -3 & -2 & 4 \\ 5 & 2 & 4 \\ -2 & 0 & -8 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1 + R_3} \begin{pmatrix} -3 & -2 & 4 \\ 0 & 0 & 0 \\ -2 & 0 & -8 \end{pmatrix} \xrightarrow{R_2 = R_3} \begin{pmatrix} -3 & -2 & 4 \\ 0 & 0 & 0 \\ -2 & 0 & -8 \end{pmatrix} \xrightarrow{R_3 = R_2} \begin{pmatrix} -3 & -2 & 4 \\ 0 & 0 & 0 \\ -2 & 0 & -8 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & -2 & 4 \\ -2 & 0 & -8 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 = R_2 + \frac{2}{3}R_1} \begin{pmatrix} -3 & -2 & 4 \\ 0 & \frac{4}{3} & \frac{16}{3} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow T = \langle (-3, -2, 4), (0, \frac{4}{3}, \frac{16}{3}) \rangle$$

$$\Rightarrow \dim \langle T \rangle = 2$$

$$\begin{aligned} -2 + \frac{2}{3} \cdot 4 &= -\frac{2}{3} \\ 0 + \frac{4}{3} &= \frac{4}{3} \\ -8 - \frac{2}{3} \cdot 4 &= -\frac{32}{3} \end{aligned} \quad \text{④}$$

$$S+T = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ -3 & -2 & 4 \\ 0 & 0 & 16 \end{pmatrix} \xrightarrow{-4R_2 + R_4 \rightarrow R_4} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ -3 & -2 & 4 \\ 0 & 0 & 16 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + 3R_1} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 0 & 16 \\ 0 & 0 & 16 \end{pmatrix}$$

$$+32 + 16 = 48$$

$$\begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 0 & 16 \\ 0 & 0 & 48 \end{pmatrix} \xrightarrow{R_3 \leftarrow 2R_2 + R_3} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 0 & 0 \\ 0 & 0 & 48 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 1 & -8 \\ 0 & 0 & 48 \end{pmatrix}$$

$$4 + 12$$

$$\Rightarrow \dim(S+T) = 3.$$

$$S+T = \langle (1, 0, 4), (0, 1, -8), (0, 0, 48) \rangle.$$

$$\dim S+T = \dim S$$

$$\dim S + \dim T = \dim(S+T) + \dim(S \cap T)$$

$$2 + 2 = 3 + \dim(S \cap T)$$

$$\Rightarrow \dim(S \cap T) = 1.$$